

Lecture 3

DISORDERS OF VESTIBULAR SYSTEM

Disorders of vestibular system cause vestibular symptoms and sign

VESTIBULAR SYMPTOMS

- 1) Internal vertigo
- 2) Dizziness (Spatial disorientation)
- 3) Vestibulo-visual symptoms
 - i) External vertigo
 - ii) Oscillopsia
 - iii) Visual lag
 - iv) Visual tilt
 - v) Movement-induced blur
- 4) Postural symptoms
 - i) Unsteadiness
 - ii) Pulsion
 - iii) Balance-related near-fall
 - iv) Balance-related fall

Table 1 Vestibular disorders

Peripheral (Lesions of end organs vestibular nerve)	Central (Lesions of brainstem and central connections)
• Ménière's disease • Benign paroxysmal positional vertigo • Vestibular neuronitis • Labyrinthitis • Vestibulotoxic drugs • Head trauma • Perilymph fistula • Syphilis • Acoustic neuroma	• Vertebrobasilar insufficiency • Posterior inferior cerebellar artery syndrome • Basilar migraine • Cerebellar disease • Multiple sclerosis • Tumours of brainstem and fourth ventricle • Epilepsy • Cervical vertigo

Disorders of the vestibular system are divided into:

- 1. Peripheral**, which involves vestibular end organs and their 1st-order neurons (i.e. the vestibular nerve). The cause lies in the internal ear or the VIIIth nerve. They are responsible for 85% of all cases of vertigo.
- 2. Central**, which involves the central nervous system after the entrance of the vestibular nerve in the brainstem and involves vestibulo-ocular, vestibulospinal, and other central nervous system pathways. [Table 1](#) lists the common causes of vertigo of peripheral and central origin.

PERIPHERAL VESTIBULAR DISORDERS

1. Ménière's disease (endolymphatic hydrops). It is characterized by vertigo, fluctuating hearing loss, tinnitus, and a sense of pressure in the involved ear. Vertigo is of sudden onset and lasts for a few minutes to 24 hours.

Cardinal symptoms of Ménière's disease are

- (i) episodic vestibular symptoms,
- (ii) fluctuating hearing loss,
- (iii) tinnitus and
- (iv) sense of fullness or pressure in the involved ear.

PATHOLOGY

The main pathology is the distension of the endolymphatic system, mainly affecting the cochlear duct (scala media) and the saccule, and to a lesser extent the utricle and semicircular canals. The dilatation of the cochlear duct is such that it may fill the scala vestibuli; there is marked bulging of Reissner's membrane, which may even herniate through the helicotrema into the apical part of scala tympani (Figure 1).

AETIOLOGY

The main pathology in Ménière's disease is distension of endolymphatic system due to increased volume of endolymph. This can result either from increased production of endolymph or its faulty absorption or both. The exact cause of Ménière's disease is not yet known. Various theories have been postulated (Figure 2).

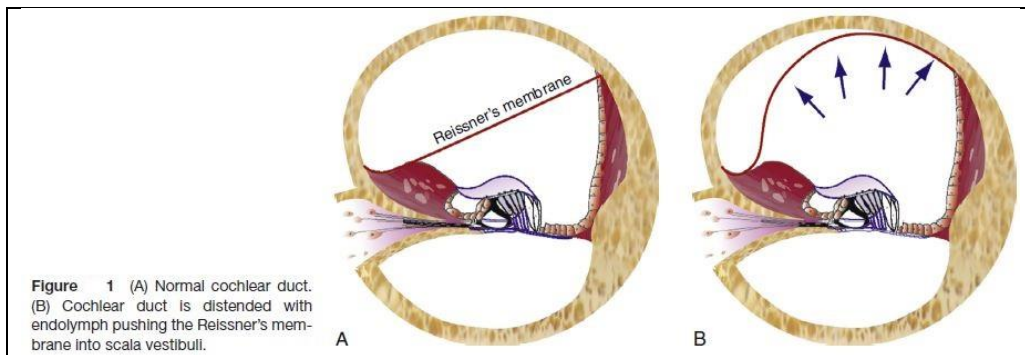


Figure 1 (A) Normal cochlear duct. (B) Cochlear duct is distended with endolymph pushing the Reissner's membrane into scala vestibuli.

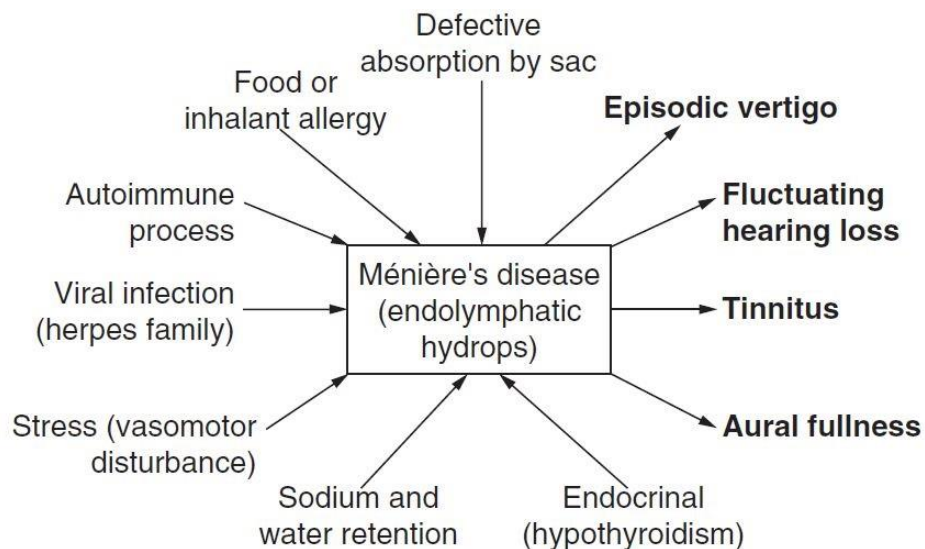


Figure 2 Aetiologic factors and symptomatology of Ménière's disease (endolymphatic hydrops).

INVESTIGATIONS

1. Pure tone audiometry. There is sensorineural hearing loss. In early stages, lower frequencies are affected, and the curve is of rising type. When higher frequencies are involved, curve becomes flat or a falling type (Figure 3).

2. Speech audiometry. Discrimination score is usually 55–85% between the attacks but discrimination ability is much impaired during and immediately following an attack.

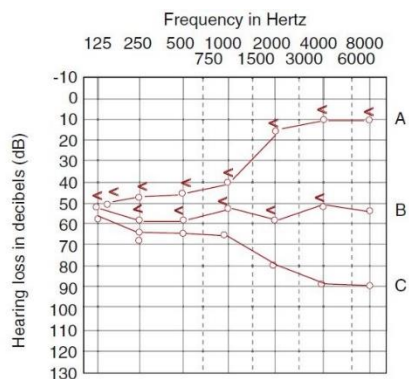


Figure 3 (A) Audiogram in early Ménière's disease. Note: Hearing loss is sensorineural and more in lower frequencies—the rising curve. As the disease progresses, middle and higher frequencies get involved and audiogram becomes flat or falling type (B & C).

TREATMENT

A. GENERAL MEASURES

1. Reassurance. Patient anxiety can be relieved by reassurance and by explaining the true nature of disease. This is particularly important in acute attack.
2. Cessation of smoking. Nicotine causes vasospasm. Smoking should be completely stopped. For some patients, this may be the only treatment necessary.
3. Low salt diet. Patients should take a salt-free diet as far as possible.
4. Avoid excessive intake of water.
5. Avoid over-indulgence in coffee, tea and alcohol.
6. Avoid stress and bring a lifestyle change. Mental relaxation exercises and yoga are helpful.
7. Avoid activities requiring good body balance. As the attack of Ménière's disease is abrupt, sometimes with no warning symptom, professions such as flying, underwater diving or working at great heights should be avoided.

B. MANAGEMENT OF ACUTE ATTACK

During the acute attack, there is severe vertigo with nausea and vomiting. Patient is apprehensive. Head movements provoke giddiness. Therefore, treatment would consist of:

1. Reassurance and psychological support to allay worry and anxiety.
2. Bed rest with head supported on pillows to prevent excessive movements.
3. Vestibular sedatives to relieve vertigo. They should be administered intramuscularly or intravenously, if vomiting precludes oral administration.

Drugs useful in acute attack are dimenhydrinate (Dramamine), promethazine chocolate (Avomine) or prochlorperazine (Stemetil). Diazepam (Valium or Calmpose) 5–10 mg may be given intravenously.

4. Vasodilators (a) Inhalation of carbogen (5% CO₂ with 95% O₂). It is a good cerebral vasodilator and improves labyrinthine circulation.

C. MANAGEMENT OF CHRONIC PHASE

When patient presents after the acute attack, the treatment consists of:

1. Vestibular sedatives. Prochlorperazine (Stemetil) 10 mg, thrice a day, orally for two months and then reduced to 5 mg thrice a day for another month.
2. Vasodilators. Betahistine (Vertin) 8–16 mg, thrice a day, given orally, also increases labyrinthine blood flow by releasing histamine in the body.
3. Diuretics. Sometimes, diuretic furosemide, 40 mg tablet, taken on alternate days with potassium.

2. Benign paroxysmal positional vertigo (BPPV). It is characterized by vertigo when the head is placed in a certain critical position. There is no hearing loss or other neurologic symptoms. Positional testing establishes the diagnosis and helps to differentiate it from positional vertigo of central origin ([Table 1](#)). Disease is caused by a disorder of posterior semicircular canal though many patients have history of head trauma and ear infection. It has been demonstrated that otoconial debris, consisting of crystals of calcium carbonate, is released from the degenerating macula of the utricle and floats freely in the endolymph. When it settles on the cupula of posterior semicircular canal in a critical head position, it causes displacement of the cupula and vertigo. The vertigo is fatiguable on assuming the same position repeatedly due to dispersal of the otoconia but can be induced again after a period of rest.

Thus, typical history and Hallpike manoeuvre establishes the diagnosis. The condition can be treated by performing *Epley's manoeuvre*. The principle of this manoeuvre is to reposition the otoconial debris from the posterior semicircular canal back into the utricle. The doctor stands behind the patient and the assistant on the side. The patient is made to sit on the table so that when he is made to lie down, his head is beyond the edge of the table as is done in Dix-Hallpike manoeuvre. His face is turned 45° to the affected side.

The manoeuvre consists of five positions:

- *Position 1.* With the head turned 45°, the patient is made to lie down in head-hanging position (Dix-Hallpike manoeuvre). It will cause vertigo and nystagmus. Wait till vertigo and nystagmus subside.
- *Position 2.* Head is now turned so that affected ear is up.
- *Position 3.* The whole body and head are now rotated away from the affected ear to a lateral recumbent position in a face-down position.
- *Position 4.* Patient is now brought to a sitting position with head still turned to the unaffected side by 45°.
- *Position 5.* The head is now turned forward and chin brought down 20°.

There should be a pause at each position till there is no nystagmus or there is slowing of nystagmus, before changing to the next position. After manoeuvre is complete, patient should maintain an upright posture for 48 h. Eighty per cent of the patients will be cured by a single manoeuvre.

If the patient remains symptomatic, the manoeuvre can be repeated. A bone vibrator placed on the mastoid bone helps to loosen the debris.

3. Vestibular neuronitis

Vestibular neuritis, also known as vestibular neuronitis, is a representative peripheral vertigo. The causes of vestibular neuritis are not yet clear, but mainly viral infection, ischemia, and immune-mediated mechanisms.

Clinical features

The characteristic clinical features of vestibular neuritis are abrupt true-whirling vertigo lasting for more than 24 hours, and no presence of cochlear symptoms and other neurological symptoms and signs.

Diagnosis

To accurately diagnose vestibular neuritis, various diagnostic tests such as the head impulse test, bithermal caloric test, and vestibular-evoked myogenic potential test are conducted.

Treatment

Symptomatic therapy, specific drug therapy, and vestibular rehabilitation therapy have been studied and implemented for the treatment of vestibular neuritis.

4. Labyrinthitis.

- *Circumscribed labyrinthitis* is seen in cases of unsafe type of chronic suppurative otitis media (CSOM) and fistula test is positive.
- *Serous labyrinthitis* is caused by trauma or infection (viral or bacterial) adjacent to inner ear but without actual invasion. There is severe vertigo and sensorineural hearing loss. A partial or full recovery of inner ear functions is possible if treated early.
- *Purulent labyrinthitis* is a complication of CSOM. There is actual bacterial invasion of inner ear with total loss of cochlear and vestibular functions. Vertigo in this condition is due to acute vestibular failure. There is severe nausea and vomiting. Nystagmus is seen to the opposite side due to destruction of the affected labyrinth.

5. Vestibulotoxic drugs. Several drugs cause ototoxicity by damaging the hair cells of the inner ear. Some primarily affect the cochlear while others affect the vestibular labyrinth.

Aminoglycoside antibiotics particularly streptomycin, gentamicin and kanamycin have been shown to affect hair cells of the crista ampullaris and to some extent those of the maculae. Certain other drugs which cause dizziness or unsteadiness are antihypertensives, labyrinthine sedatives, oestrogen preparations, diuretics, antimicrobials (nalidixic acid, metronidazole) and antimalarials. However, their mode of action may be different.

6. Head trauma. Head injury may cause concussion of labyrinth, completely disrupt the bony labyrinth or VIIIth nerve, or cause a perilymph fistula. Severe acoustic trauma, such as that caused by an explosion, can also disturb the vestibular end organ (otoliths) and result in vertigo.

7. Perilymph fistula. In this condition, perilymph leaks into the middle ear through the oval or round window. It can follow as a complication of stapedectomy, or ear surgery when stapes is accidentally dislocated. It can also result from sudden pressure changes in the middle ear (e.g. barotrauma, diving, forceful Valsalva) or raised intracranial pressure (weightlifting or vigorous coughing). A perilymph fistula causes intermittent vertigo and fluctuating sensorineural hearing loss, sometimes with tinnitus and sense of fullness in the ear (*compare* Ménière's disease).

8. Syphilis. Syphilis of inner ear, both acquired and congenital, causes dizziness in addition to sensorineural hearing loss. Late congenital syphilis usually manifesting between 8 and 20 years, mimics Ménière's disease with episodes of acute vertigo, sensorineural hearing loss and tinnitus. Hennebert's sign, i.e. a positive fistula test in the presence of an intact tympanic membrane, is present in congenital syphilis. Neurosyphilis (tertiary acquired) can cause central type of vestibular dysfunction.

9. Acoustic neuroma. It has been classified in peripheral vestibular disorders as it arises from CN VIII within internal acoustic meatus. It causes only unsteadiness or vague sensation of motion. Severe episodic vertigo, as seen in the end organ disease, is usually missing. Other tumours of temporal bone (e.g. glomus tumour, carcinoma of external or middle ear and secondaries), destroy the labyrinth directly and cause vertigo.

VESTIBULAR MIGRAINE

Vestibular migraine is now the internationally accepted term for a type of migraine that mainly presents with vestibular symptoms. They may occur without headaches, although in around half of the patients are followed by or occur with a headache or visual aura.

According to the purposed diagnostic criteria for vestibular migraine, the diagnosis of vestibular migraine is based on the presence of the following conditions:

- 1) recurrent vestibular symptoms,
- 2) a current or previous history of migraine,
- 3) a temporal correlation between those vestibular symptoms and migraine attacks, and
- 4) exclusion of another cause of the vestibular symptoms.

Pathophysiology of vestibular migraine

Due to cortical spreading depression

- 1) Cerebral vasospasm (AICA and internal auditory artery)
- 2) This hypoxic event is thought to be the cause for the aura
- 3) Cerebral vasorelaxation - Headache

Clinical characteristics

Symptoms

For most patients, vestibular migraine is an episodic disorder; however, the duration of attacks ranges from seconds to days. Vestibular migraine has a strong female predominance of up to 5 to 1,11 and vestibular migraine often begins several years after typical migraine.

Some patients can have a headache-free interval of several years before the onset of vestibular migraine. Vestibular migraine might begin in place of headache, especially in perimenopausal women.¹⁹ Vestibular migraine is more common in patients without aura than in patients with aura. The temporal association

between the vestibular symptoms and migrainous symptoms such as headache is quite variable between patients and the association might be inconsistent in an individual.

Treatment

The treatment of migraine is often a multifaceted approach, involving abortive therapies, preventative medications, and behavioral changes.

Abortive Medications

The goals of acute or abortive treatment of migraine headache are to relieve pain and to restore function as quickly as possible. As a general rule, abortive medications are most effective when taken early in the course of the headache.

Patients with mild to moderate migraine headaches may be adequately treated with simple analgesics such as ibuprofen or naproxen, but those whose headaches are typically moderate to severe should be started on a more “aggressive” therapy, such as triptans or ergotamines.

Preventive Medication

When patients suffer with frequent migraine headaches, they may become candidates for the use of preventive medication options (Table 2).

Table 2 Preventive medication options for prevention of migraine and level of evidence

Preventive medication	Common dosages (mg)	Level of evidence
Divalproex sodium/sodium valproate ^a	250–1,500	A
Topiramate ^a	50–200	A
Metoprolol	50–150	A
Propranolol ^a	80–240	A
Timolol ^a	20–60	A
Amitriptyline	10–150	B
Venlafaxine	75–225	B
Atenolol	50–200	B
Nadolol	20–160	B
Lisinopril	10–20	C
Candesartan	16	C

Adapted from Silberstein et al. [12]

A=established efficacy (2 or more class 1 trials)

B=probably efficacy (1 class 1 trial or 2 class 2 studies)

C=probably efficacy (1 class 2 study)

^aFDA approval